



MIDDLE EAST TECHNICAL UNIVERSITY
NORTHERN CYPRUS CAMPUS

DEPARTMENT
OF
COMPUTER ENGINEERING

CNG 351
Data Management and File Structures
Assignment 3
(6% of actual grade)

DUE DATE: 05 December, Thursday, 23:55 (Cyprus Time).

PURPOSE

This assignment aims to help you revise the following two topics: **Normalisation** and also **Relational Algebra**. Therefore, this assignment has two main parts. In the first part, you are given logical designs of sample relations in a given database and you are asked to normalise them. In the second part, you are given a database and you are asked to answer some queries via relational algebra.

IMPORTANT RULES

1. Please make sure that your solutions are clearly explained.
2. Please make sure that your normalisation process is justified with relevant functional dependencies and explained clearly.
3. When you do normalisation, please show all your steps.
4. Create a PDF file for your solution, and upload only one PDF file to ODTUClass, one team member is enough to upload the solution.
5. Assignments will be completed by a team of two people that was formed for the previous assignment. if there is a problem with your partner, please let us know.
6. Please submit a report that includes a cover page with the team details including their names/surnames and also student IDs.

GRADING

This assignment has two parts and the overall grading will be as follows:

1. Part 1: Use Case 1: 25 points;
2. Part 1: Use Case 2: 30 points;
3. Part 2: 40 points;
4. Report: 5 points. A good report means type written, complete (every section fulfilled), with clear explanations in English (where relevant), and submitted via ODTUCLASS as one combined PDF document. The first page of the report must be a title page which should clearly state team details and assignment number. Each use case should also include Assumptions clearly written. The footer of all subsequent pages should be numbered in the format x of y (eg 2 of 6), etc.

PART 1: NORMALISATION

In this part, you are given two logical database designs and you are asked to normalise them. Please, check the following against the different normalization rules (i.e., 1NF, 2NF, 3NF). Fix the schema if any of the normalization rules is violated.

Use Case 1 – Online Tutoring System (25 pts)

In Cyprus, a smart online tutoring system was developed to address the growing demand for students for flexible and additional academic support offered by tutors who specialize in various subjects. This platform connects students with the tutors in a way that allows them to conduct private tutoring sessions either for supplementary help in their ongoing courses or to gain proficiency in a new subject. The system integrates functionalities other than connecting students with tutors, such as progress tracking, course and lecture management, and secure payment processing. A database designer has already designed the database with the following relation. You need to show the steps required to normalise this relation to 3NF.

RelationR(student_id, student_phone, student_university, tutor_id, tutor_rating, payment_id, payment_amount, payment_date, lecture_start_time, course_id, course_name, admin_id, admin_names)

FD₁ : {student_id} – > {student_phone, student_university}

FD₂ : {admin_id} – > {admin_name}

FD₃ : {tutor_id} – > {tutor_rating, admin_id}

FD₄ : {lecture_start_time, course_id} – > {lecture_cost, lecture_end_time, tutor_id}

FD₅ : {course_id} – > {course_name}

FD₆ : {payment_id} – > {payment_amount, payment_date, lecture_start_time, course_id}

FD₇ : {student_id, lecture_start_time, course_id} – > {payment_id}

Use Case 2 – Desire for Velocity (30 pts)

Desire For Velocity is a multiplayer competitive racing game developed by some brilliant indie game developers in METU NCC. It's a real-time racing game with different cars and car classes, races with matchmaking, tournaments, and single-player drift mode. Table ?? gives the data in one of the relations created. For this relation, you must check the data and create functional dependencies. Once you create functional dependencies, you must normalise the relation according to 3NF

and BCNF. Please explain your functional dependencies and then show the steps required to normalise this relation. Please note the following about the attributes:

player_id: Every player in the system has a unique ID. With a player_id, the system stores a nickname and elo_score.

nickname: Each player has a single nickname stored in the system.

elo_score: Elo score or Elo ranking is a value for ranking player skill. Higher elo means better skills and a better player. Each player has an Elo score stored in the system.

car_name: This holds the name of the car. Please note that the players can have multiple cars. Each car has a single name in the system. A player cannot have two cars with the same name, but different players can have the same car name.

car_brand: The system stores the brand of a given car.

car_model: The system stores the model of a given car.

race_id: The system also records the race details, and each race is identified with a unique ID.

laps: It stores the number of laps of a given race identified with the race_id.

map: It stores the map details of a given race identified with the race_id.

position: A car can participate in multiple races, and the position holds the position information of a car in a given race.

t_id: This is tournament id. Every tournament is uniquely identified with an ID. A tournament can have many races involved.

size: This shows the size of a tournament.

t_winner: This is tournament winner. This shows the player (identified by an ID) who is the tournament winner. A tournament has one winning player but ofcourse a player can win many tournaments.

c_id: This is the cosmetics id. A car can have many cosmetics added to it. Each of these cosmetics is identified with a unique id.

rarity: Each cosmetic unique item has a rarity value.

player_id	nickname	e_score	car_name	car_brand	car_model	raceID	laps	map	position	t_id	size	t_winner	c_id	rarity
2981	snakeserdar	1321	lokмата	Anadol	A5	216	1	Alpine Ascent	5	103	6	5252	20218	Common
2981	snakeserdar	1321	lokмата	Anadol	A5	573	2	Tokyo Driftways	1	101	5	2981	20218	Common
1351	TylerDurden	1169	SoapMachine	Chevrolet	Camaro	848	2	Monaco Circuit	2	101	5	2981	49205	Uncommon
1351	TylerDurden	1169	SoapMachine	Chevrolet	Camaro	526	3	Arctic Edge	3	105	7	2424	49205	Uncommon
1351	TylerDurden	1169	doNorT4lk	Suzuki	Swift RS	482	3	Istanbul Inter	2	105	7	2424	12151	Uncommon
5252	kharp haz	959	McQueen	Chevrolet	Corvette	216	1	Alpine Ascent	1	103	6	5252	14300	Rare
5252	kharp haz	959	scubaRacer	BMW	1.16	482	3	Istanbul Inter	4	105	7	2424	56980	Common
3459	LadyDeath	803	McQueen	Subaru	BRZ	526	3	Arctic Edge	7	105	7	2424	59143	Uncommon
2424	DominicForetto	1520	FamilyFirst	Dodge	Charger	431	2	Sahara Speedway	2	105	7	2424	83906	Common
2424	DominicForetto	1520	FamilyFirst	Dodge	Charger	526	3	Arctic Edge	1	105	7	2424	83906	Common
3153	SametBaba	1046	kolpha	Tesla	Roadster	216	1	Alpine Ascent	3	103	6	5252	71338	Rare
3153	SametBaba	1046	kolpha	Tesla	Roadster	106	3	Rio Rally	1	103	6	5252	71338	Rare
3153	SametBaba	1046	DocHudson	Hudson	Hornet	482	3	Istanbul Inter	3	105	7	2424	31263	Common
7045	xKralTR	1091	Kha-Rhambhol	Audi	R8	848	2	Monaco Circuit	4	101	5	2981	97072	Rare
4019	yesilady	1167	Cardinality	Mazda	Miata	106	3	Rio Rally	2	103	6	5252	48190	Common
4019	yesilady	1167	Cardinality	Mazda	Miata	681	3	Death Valley Run	1	102	5	1351	48190	Common
2147	Sp33d4dd1ct	1428	GottaGoF4st	Bugatti	Tourbillon	431	2	Sahara Speedway	8	102	5	1351	56897	Common

PART 2 – RELATIONAL ALGEBRA

METU NetRegister (40 pts)

At METU NCC, students must register their devices to our campus network registration system to gain access to the campus network. The university has implemented a comprehensive network registration system called METU NetRegister to manage device registrations and facilitate this process. The main goal of this application is to store information about students, devices, registrations, network engineers, and related network issues and keep track of the registered devices' VPN connection details. Based on the requirements given in the previous assignment, a database was designed and mapped to a logical design as follows. Please note that this is not the direct mapping of the requirements in the previous assignment. To address the following queries, you must write some **Relational Algebra** queries. Please note that this is not the full design of the requirements in the previous step.

Relations

Student

studentID, name, surname, email, dob, departmentCode

Device

macAddress, model, os, type, owner [FK: Student: studentID]

DeviceSpecification

macAddress [FK: Device: macAddress], specification

Registration

regID, timestamp, status, type, deviceMacAddress [FK: Device: macAddress], approvedBy [FK: NetworkEngineer: engID]

NetworkEngineer

engID, name, surname, dob

BlockHistory

regID [FK: Registration: regID], startTimestamp, duration, description, blockedBy [FK: NetworkEngineer: engID]

VPNConnectionHistory

regID [FK: Registration: regID], startTimestamp, duration

NetworkIssues

regID [FK: Registration: regID], timestamp, description, comments, resolvedBy [FK: NetworkEngineer: engID]

Domains

Student.department_code = 355, 356, 370, 378

Device.mac_address = A 12-character alphanumeric attribute that is used to identify individual electronic devices on a network. For example, 00-B0-D0-63-C2-26 is stored in our DB As 00B0D063C226

Device.type = mobile, laptop

Registration.status = approved, declined

Registration.type = wireless, wired

Timestamp is stored in the system as “yyyy-mm-dd hh:mm:ss”, e.g., 2024-11-11 18:00:00

Society.type = Art, Sport and Technical

BlockHistory.duration = It shows the number of days for blocking

VPNconnectionHistory.duration = It shows the number of hours of connection

Queries

1. List the model and operating system (os) of all devices stored in the system.
2. List the Mac addresses of all the approved registrations after 11/11/2024.
3. List the name and surname of the students who own a laptop recorded as a device in the system.
4. List the name and surname of the students whose device registration has declined since the beginning of this year. You can take the beginning of the year as 1/1/2024.
5. A student claims that a network engineer regularly blocks his registered device. We now need to write a query to see if that is correct. We need to identify all the block cases for a student called “Ahmed Zorlu” done by “Mehmet Muhendis”. Your report needs to show the registration ID, startTimestamp, duration and description for each block case.
6. Our IT manager would like to see a report that shows the total duration of the blocked registrations for wireless and wired registration types. The report will include the total duration for wireless and the total duration for wired blockage.